MS: BST

Special class

Love Babbar • Aug 3, 2025

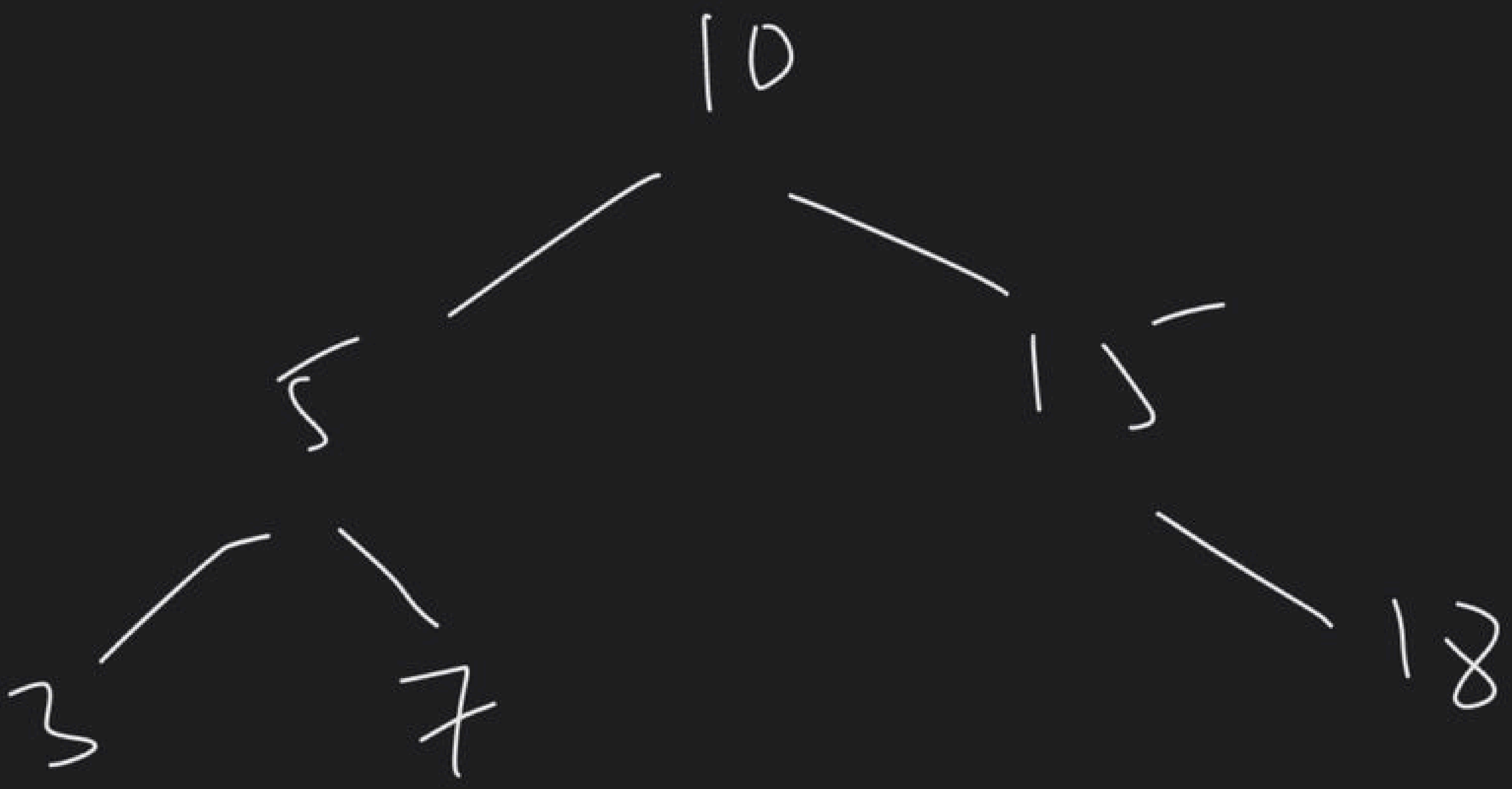
① Balanced Binary Search tree

⇒ LC-108 Sorted array to BST

I) BST → inorder → Sorted array

II Re-build → Balanced BST

② Range Sum of BST



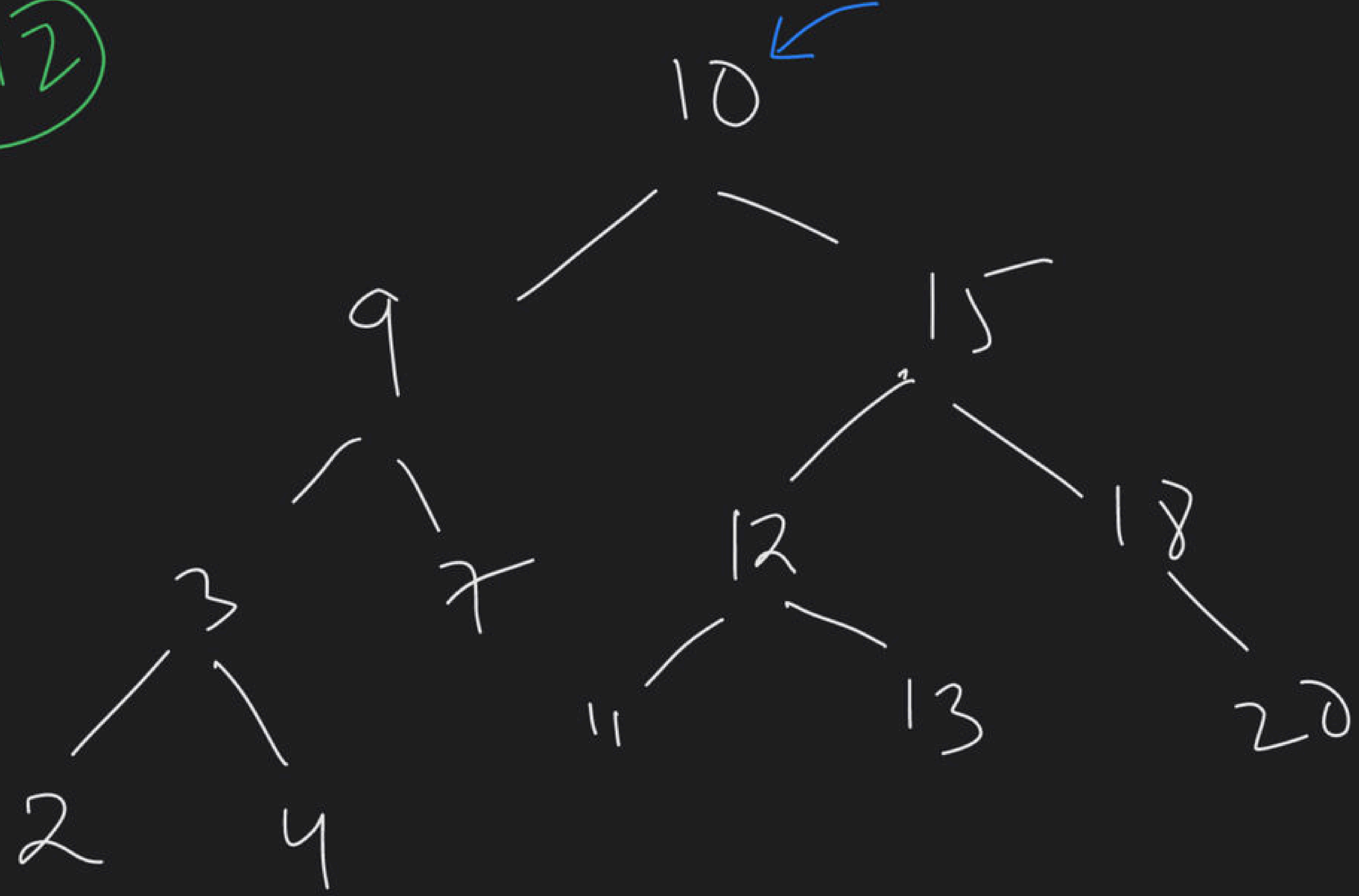
[l, h]

[7, 15]

TC $O(n)$ S.C $O(h)$

M1 Traverse whole tree & apply the cond^n

M2



[7, 15]

```
csum = 0
if (withinRange)
{
csum += root->val
      + Re(left)
      + Re(right)
}
```

- Pruning

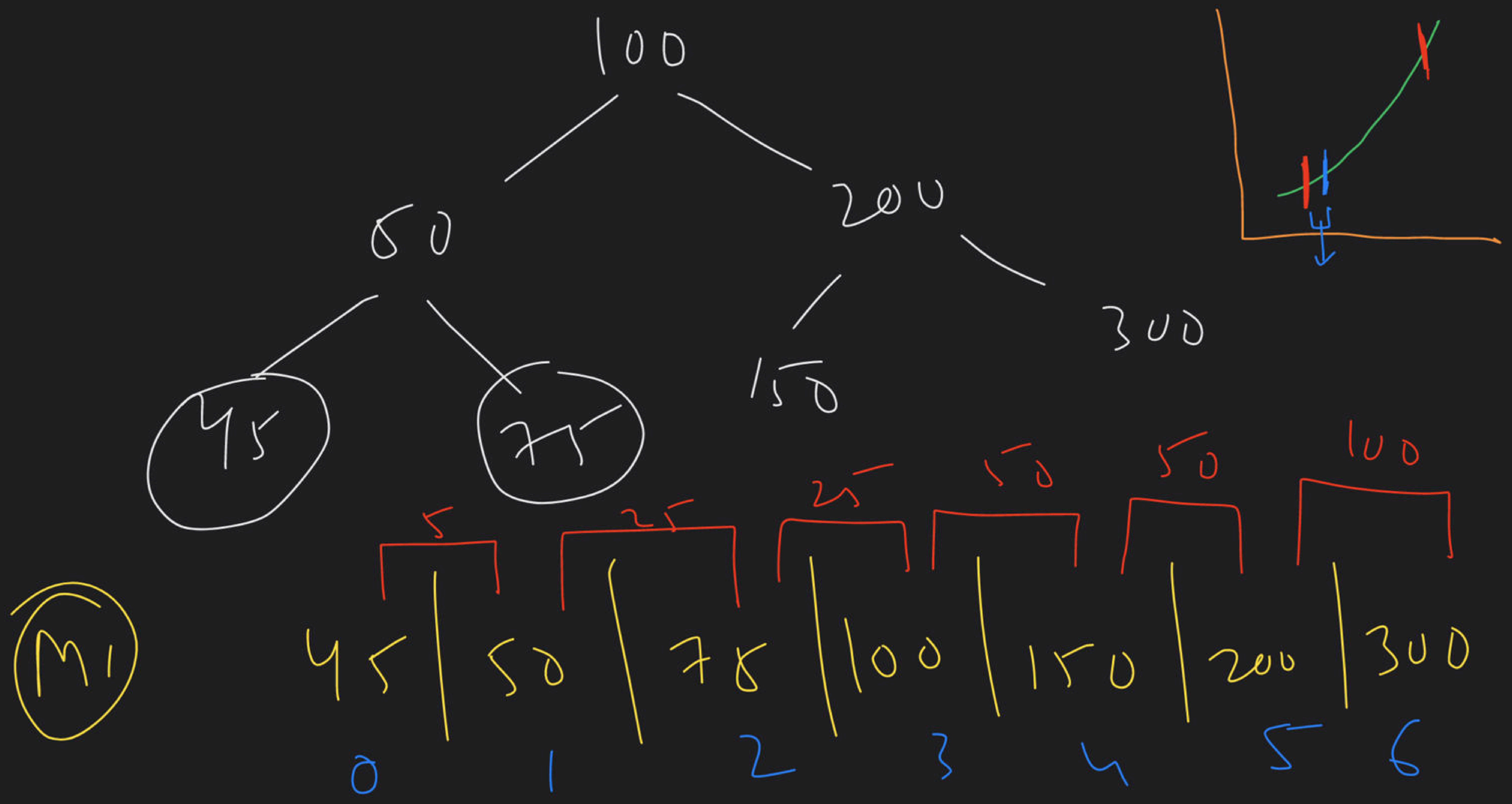
```
else {   // out of range

    if (root->val < low)
    {
        csum += RE(root->right)
    }
    else (root->val > high)
    { csum += RE(root->left)

    }
```

③ LC 783: Min Distance b/w
BST Nodes

9:35 PM Think & Read

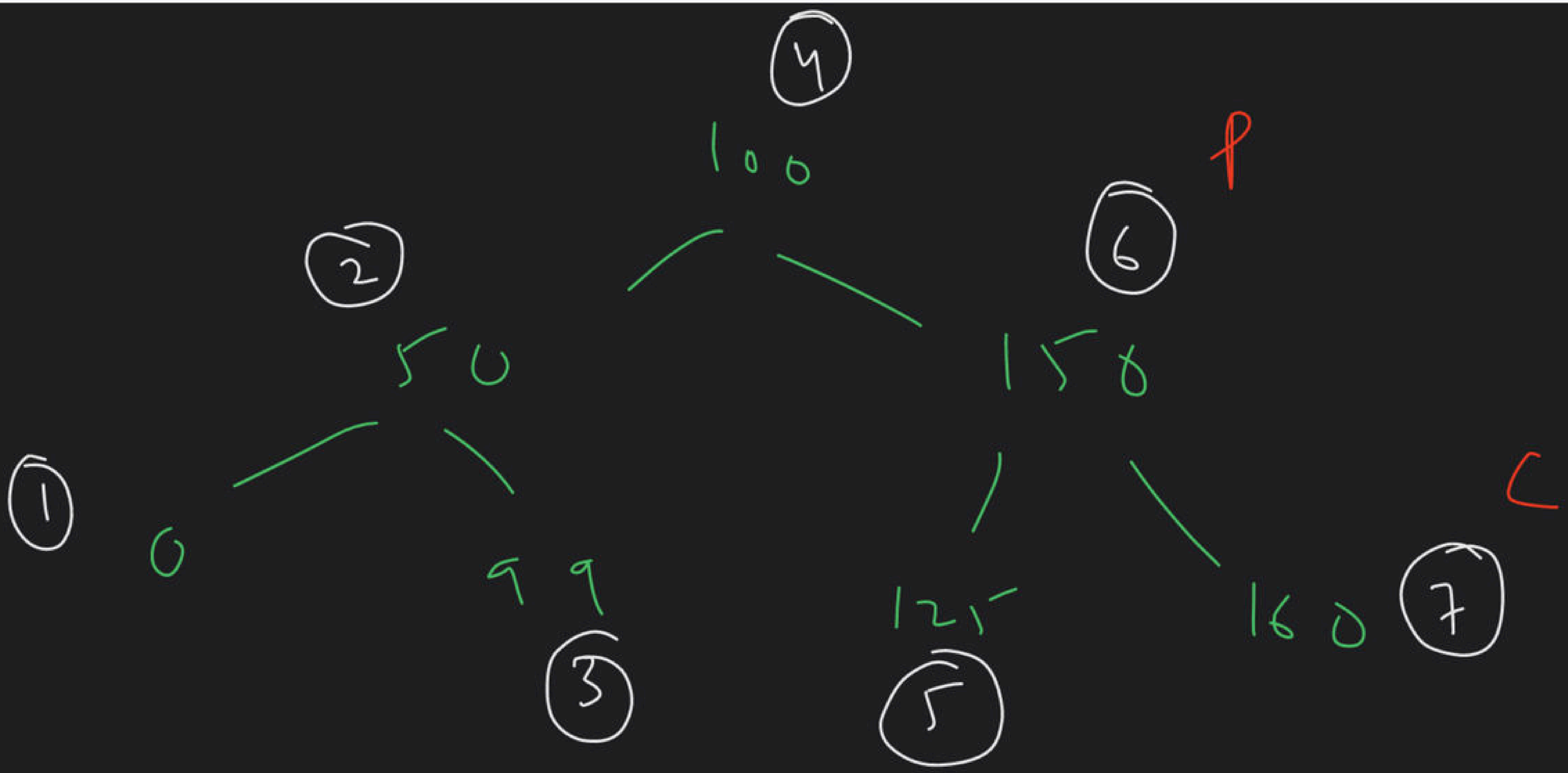

100
50
200
45
75
150
300
5
25
25
50
50
100
M1
45
50
78
100
150
200
300
0
1
2
3
4
5
6

1. inorder $O(n)$

2. store $O(n)$ space

3. Vector traverse → $O(n)$

M2 Avoid using extra vector space?



$\min = \cancel{0}\ \cancel{50}\ \cancel{49}\ \boxed{1}$

4 LC 98 → Validate BST

5 min to T&R

9:55 pm

(M1) BST → inorder → Sorted

BT → inorder → vector

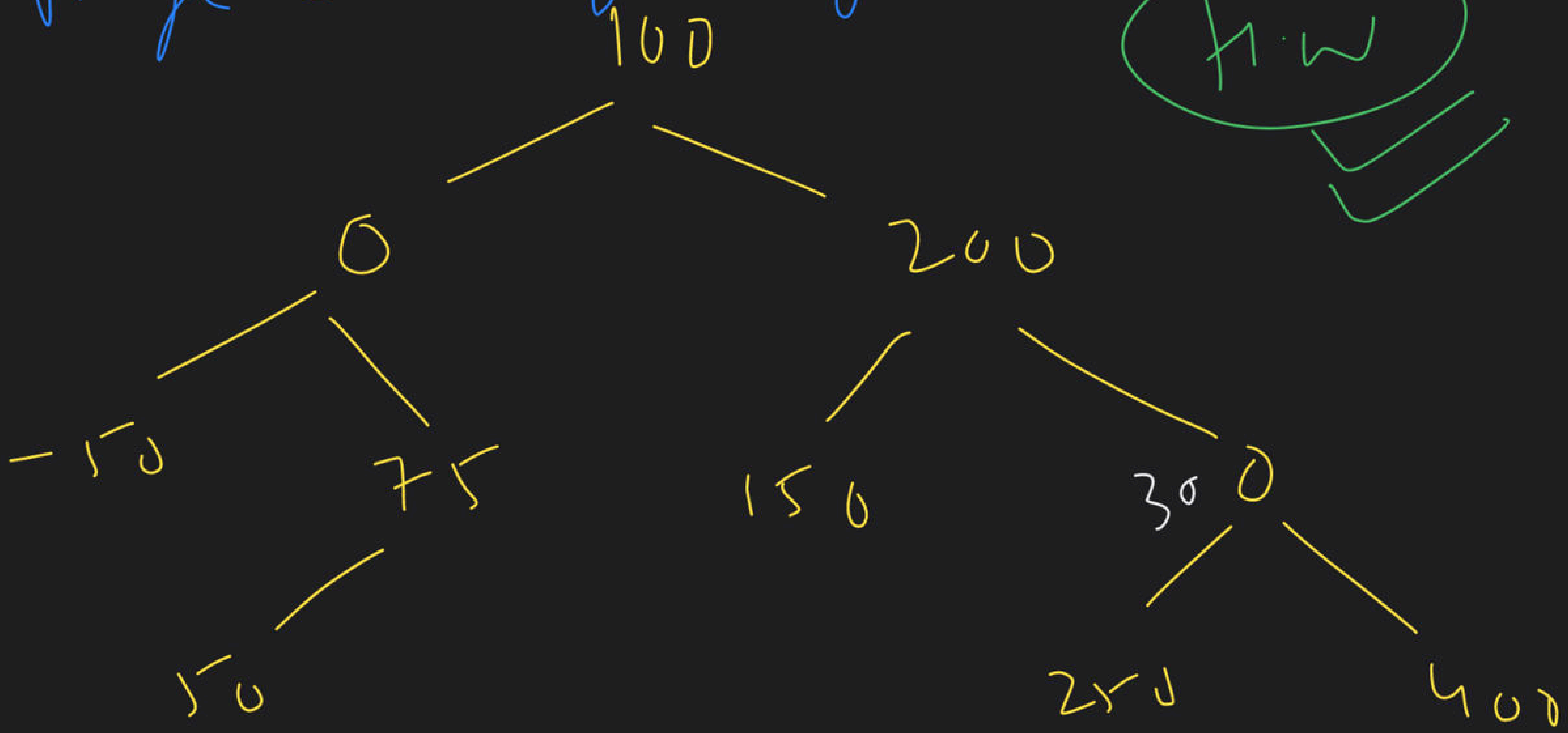
↓

Check sorted?

BST ✓ ← → ✗ Not Bst

M2 → Range checking using Postorder

H.W

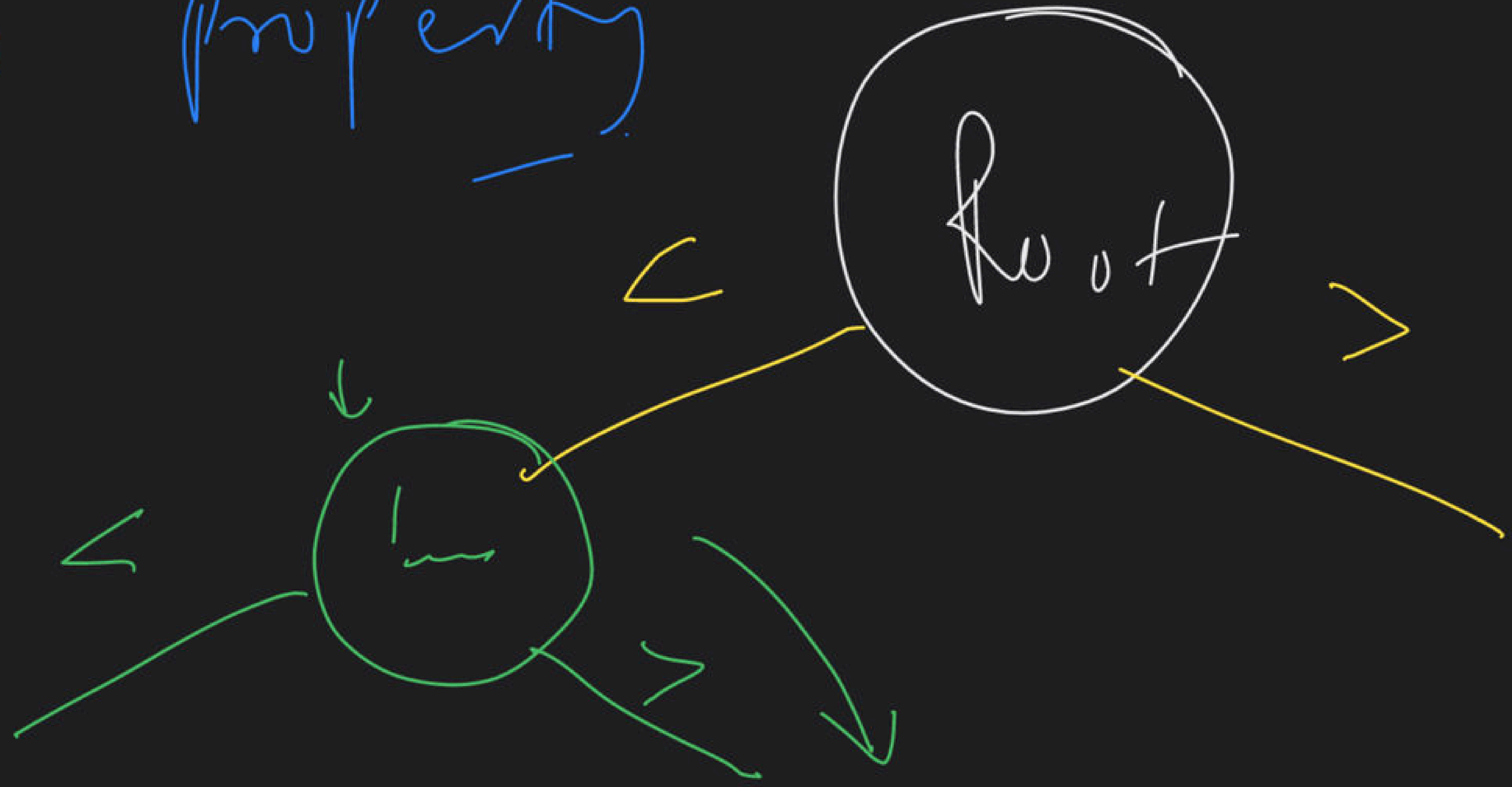


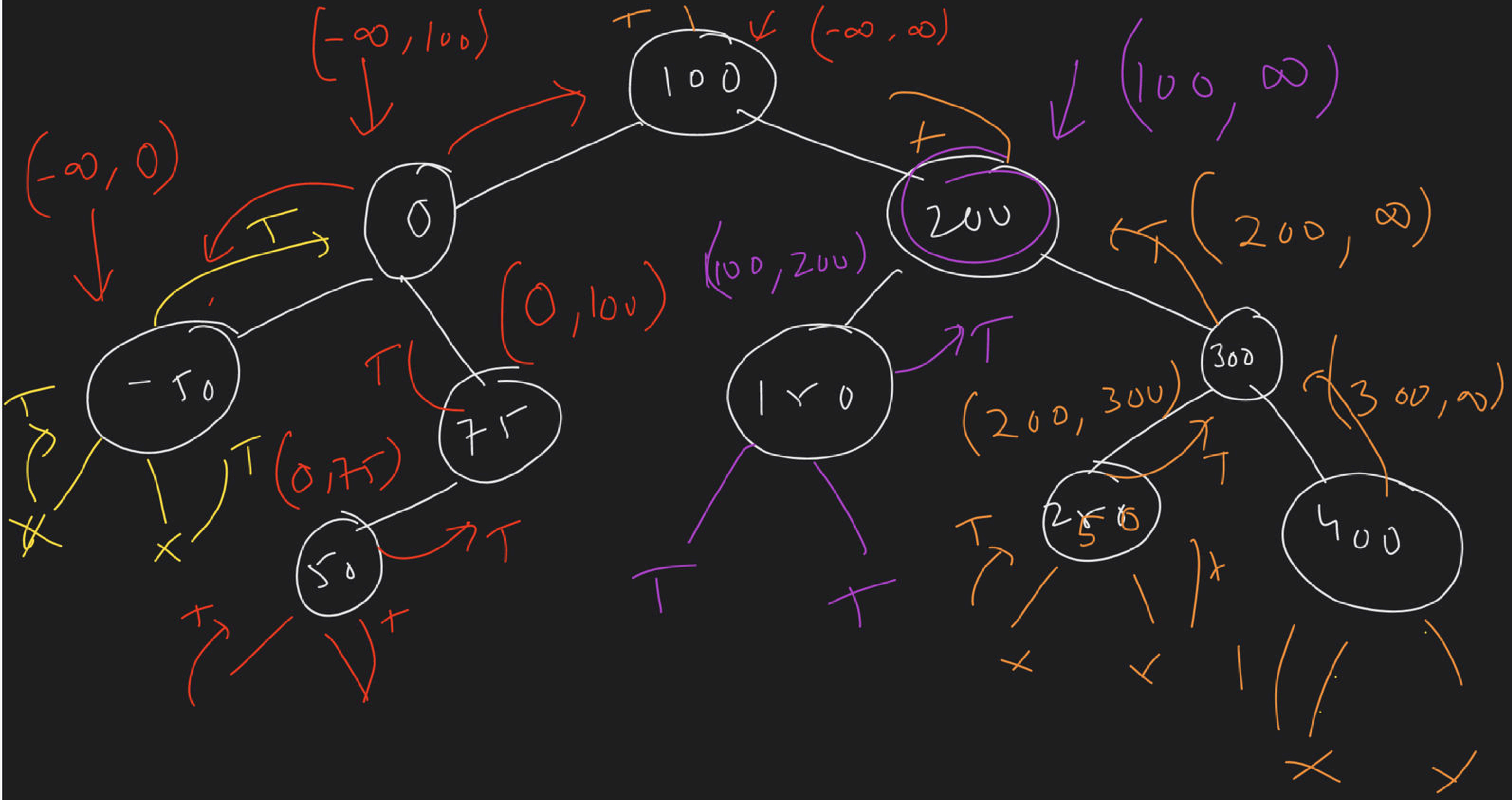
Post order traversal → Max. Sum BST

↳ LC 1373

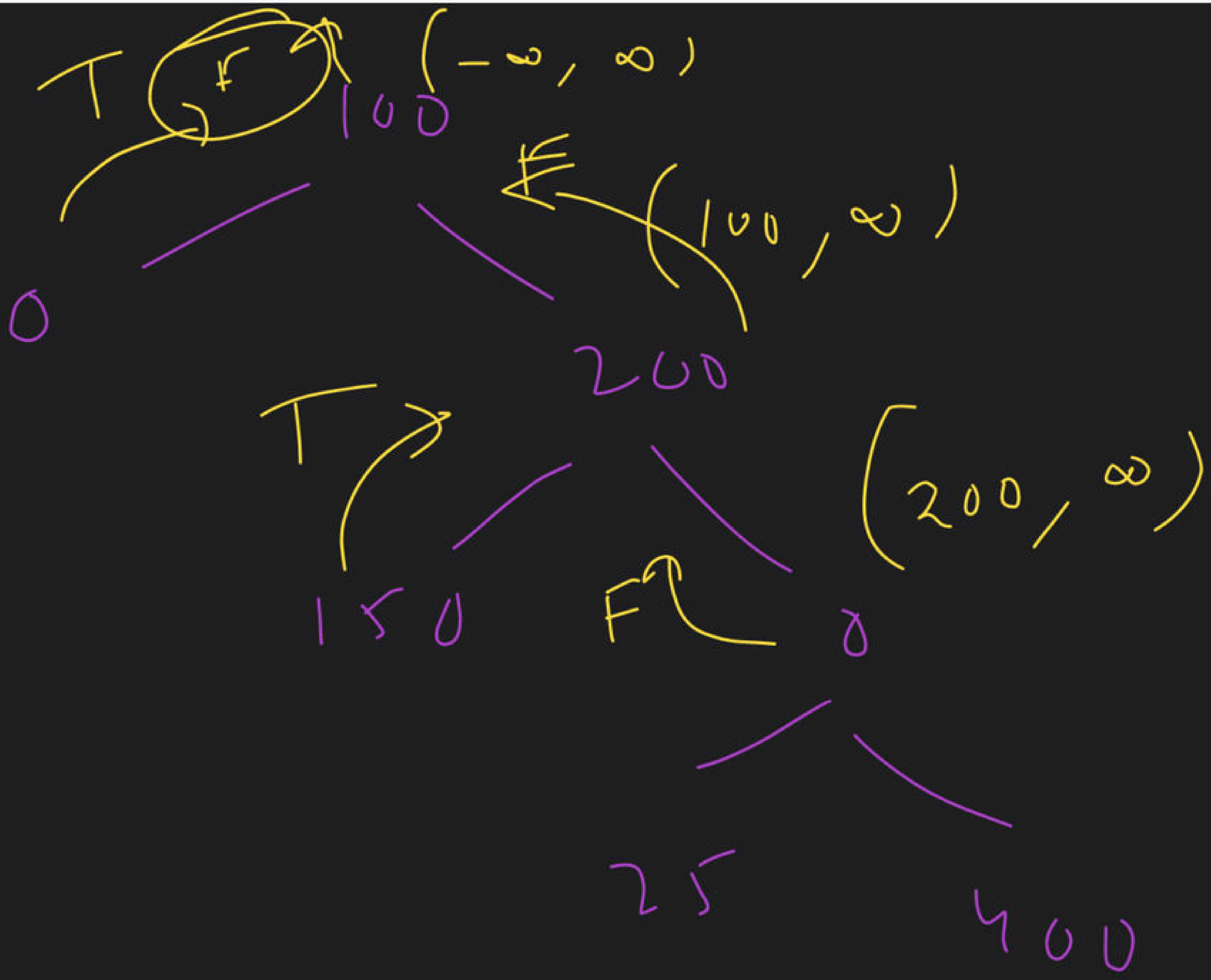
M3 Range checking using Preorder

→ BST property





(-∞, ∞)
100
(-∞, 100)
(100, ∞)
0
200
(-∞, 0)
(0, 100)
(100, 200)
(200, ∞)
-50
75
150
300
(0, 75)
(200, 300)
(300, ∞)
50
250
400
T



100
(-∞, ∞)
T
F
0
F
(100, ∞)
200
T
150
F
0
(200, ∞)
25
400

5 LC173 BST Iterator

Try

6 LC 99: Recover BST

10:30pm TfR

M1 B.F

1 Identify the Violating nodes

Vector → Inorder

BST store

BST Populate

10	20	30	60	50	40	70	80	90
0	1	2	3	4	5	6	7	8

↓ sort

10	20	30	40	50	60	70	80	90

1. vector store → inorder
2. sort → nlogn
3. Inorder → refill value

M2

Store X → Traverse kuch kuch
solve ??

1 Inorder Normal traversal

↳ Live inorder solve

10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 | 90 | 100

violation case

1. adjacent nodes → swapped

2. Non-adjacent nodes → swapped

① adjcent node

10 | 20 | 30 | 40 | 60 | 50 | 70 | 80 | 90 | 100

P C

violating Node

First = 60

second = 50

{ if (C→val < P→val)

//violation case

if (F == Null) First = prev;

second = Current;

}

② Non-adjant

10 | 20 | 30 | 40 | ~~90~~ 50 | 60 | 70 | 80 | ~~50~~ 90 | 100

After finding violating Node.

We can swap

violating

F = 90

S = ~~60~~ 50

TC → O(n)

SC → O(n)

M3 O(1)

Inorder → RE

Morris